DVGW G 625 · 4 PA METHOD

Wood Stove Field Differential-Pressure Test Procedure

Basis: DVGW Arbeitsblatt G 625, the "4/8 Pa-Test" — the field-verification method a German Schornsteinfeger (chimney sweep) performs when a combustion appliance shares an air volume with mechanical ventilation/exhaust equipment.

Purpose: demonstrate, by direct field measurement rather than manufacturer paperwork alone, that under worst-case operating conditions (all air-extracting devices at maximum power, fire lit) the pressure differential between the appliance's room and outdoors stays under the applicable threshold with any make-up air (MUA) interlock engaged, and — where an interlock exists — to show by contrast that the threshold is exceeded with it disabled. Use this to (a) confirm a CO-protection interlock functions as designed, and (b) support a PHI submission or exemption request for a wood stove whose room-sealed/RLU status is unconfirmed or in question.

Pass threshold: 4 Pa by default. Use the stricter 4 Pa (room-air-dependent / raumluftabhängig) threshold unless the appliance carries a documented DIBt Z-number or equivalent manufacturer test certificate establishing RLU (room-air-independent/raumluftunabhängig) status with test values. Pressure testing is not required with DIBt-listed appliances providing a CO detector is present in the room.

Dampers: identify every damper on the appliance's flue and combustion-air duct, and whether each is manual or automatic. Manual dampers must be opened by hand before the fire is lit.

1. Pre-test administrative

- [ ] Schedule with the blower-door tester and mechanical contractor.
- [ ] Confirm construction status: envelope complete, final airtightness (n50) test on file, stove hearth/duct/damper fully installed, MUA unit and interlock (if any) fully commissioned and functioning
- [ ] Assemble equipment: calibrated micromanometer / differential-pressure gauge (e.g. DG-700 or equivalent blower-door-grade instrument) with two static-pressure tubes, tape measure, camera, stopwatch, handheld CO meter (if available), this checklist, printed data log (§6)

2. Systems to identify and confirm functional before testing

- [ ] **Wood stove:** manufacturer/model _____ — room: _____ . Confirm dedicated combustion-air duct is unobstructed. Combustion-air damper: ☐ none ☐ manual

☐ automatic (controlled by _____). Flue damper: ☐ none ☐ manual ☐ automatic. Confirm each damper operates freely.

- ☐ **Range hood:** model _____ — confirm max/boost CFM setting and how to activate it.
- ☐ **ERV/HRV:** model _____ — confirm boost setting and how to activate it.
- ☐ **Make-up air (MUA) unit (if present):** model _____ — confirm interlock trigger (pressure switch and/or CO/CT sensor) and confirm a safe manual-override/disable method for the Condition 5 failure-mode check below.
- ☐ **CO detectors:** confirm installed per the submitted location plan and functioning.
- ☐ **Other exhaust devices** sharing this pressure zone (bath fans, dryer, other stoves, etc.) — list:

_____

_____

3. Test setup (per G625 "most unfavorable condition")

- ☐ Close all exterior windows and doors of the dwelling unit.
- ☐ Close interior doors to the stove's room to isolate it as its own pressure zone, matching normal occupant use. Note the configuration used; if time allows, repeat key readings with the door open as a secondary, less-conservative data point.
- ☐ Establish a reference outdoor pressure tap (tube run to outside air, or through the blower-door frame if used as the reference).
- ☐ With all devices off, doors/windows closed, and the stove not lit, zero/record the baseline differential pressure before starting.
- ☐ Open any manual flue and combustion-air dampers fully before lighting the fire (Condition 2 onward), as the occupant would in normal use. Record their positions.

4. Test sequence and readings

Run each condition in order. Allow the reading to stabilize (~30–60 seconds) before recording. Record the **magnitude of negative pressure** in the stove room relative to outside (Pa).

#	CONDITION	STOVE	HOOD	ERV	MUA / INTERLOCK	READING (PA)	≤ 4 PA?
1	Baseline	Off	Off	Normal	Auto (not triggered)		
2	Fire lit, otherwise baseline	Lit	Off	Normal	Auto		
3	Fire lit + hood at max/boost	Lit	Max	Normal	Auto		
4	Fire lit + hood max + ERV boost	Lit	Max	Boost	Auto		
5	Worst case, interlock manually disabled (failure-mode check)	Lit	Max	Boost	Disabled		*(expected to fail — see note)*
6	Worst case, interlock engaged as designed	Lit	Max	Boost	Engaged		

Notes: - **Condition 5 is deliberate.** It's expected to exceed 4 Pa, which demonstrates that the MUA interlock is required to keep the room within the limit. Keep this condition brief, monitor CO continuously, and treat it as a controlled check, not sustained operation. Omit this row if the installation has no interlock to test. - **Condition 6 is the condition that matters for the compliance submission** — it's the as-designed, as-occupied operating state. - If a realistic standalone occupant scenario isn't covered above (e.g., hood alone without ERV boost), add it as an extra row.

5. CO verification (concurrent with §4)

- [ ] Confirm installed CO detector(s) show no alarm through every test condition.
- [ ] If a handheld CO meter is available, log ambient CO (ppm) in the stove room at each condition alongside the pressure reading.

6. Data log

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Project: _____________
Date: _____________   Tested by: _____________   Witnessed by: _____________
Outdoor temp / wind conditions: _____________
Gauge make / model / serial: _____________   Calibration date: _____________

# | Condition                                | Reading (Pa) | CO (ppm) | Notes
1 | Baseline                                 |              |          |
2 | Fire lit, baseline                       |              |          |
3 | Fire lit + hood max                      |              |          |
4 | Fire lit + hood max + ERV boost          |              |          |
5 | Worst case, interlock disabled           |              |          |
6 | Worst case, interlock engaged            |              |          |
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7. Photographic documentation (for the compliance submission package)

- [ ] Gauge setup / pressure-tap locations
- [ ] Stove with fire lit during test
- [ ] Hood and ERV controls at max-power settings during readings
- [ ] MUA unit / interlock control panel — both engaged and manually-disabled states (if applicable)
- [ ] Rating plate of the installed appliance
- [ ] Combustion-air duct connection to the appliance, pre-panel/pre-finish if possible
- [ ] Any manual dampers in the open position

8. Outcome

- [ ] **Condition 6 reads at or below 4 Pa** → pass. Compile this procedure, the data log, and photos for the record, and include them in the owners' fireplace documentation booklet.
- [ ] **Condition 6 exceeds 4 Pa** → fail; the issue remains open.

9. After the test

- [ ] Circulate the data log and all photos from §7 to the team.